



# Ocular malignancies treated with iodine-125 low dose rate (LDR) brachytherapy at a single high-volume institution: A retrospective review

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## ABSTRACT

The aim of our study is to document our cases of choroidal melanoma treated with low dose rate (LDR) brachytherapy and to correlate the dosimetry and radiobiology with clinical effects and oncologic outcomes. Data from 157 patients treated from 2014 to 2018 with LDR brachytherapy were used for this investigation. Treatments used a collaborative ocular melanoma study eye plaque and Iodine-125 radioactive seeds. The seeds activities were chosen to deliver 85 Gy to the tumor apex or to a prescription point (if the apex < 5 mm). The plaque sizes used were 10, 12, 14, 16, 18, 20, and 22 mm including notched or deep notched. The plaques were modeled in Varian BrachyVision version 11.6 (Varian Medical Systems) with seed coordinates from the AAPM Task Group 129. The Task Group 43 from AAPM was used for brachytherapy dose planning. Dose data were extracted for the apex, prescription point, sclera, retina opposite to the implant, lens, macula, and optic disc. The radiobiological dosimetry were calculated using appropriate  $\alpha/\beta$  ratios found in the literature and then correlated to clinical side effects. Average biologically effective dose for associated organs at risk were calculated in cases where toxicity occurred. These included: radiation cataract (70.66 Gy), disc atrophy (475.49 Gy), foveal atrophy (263.07 Gy), radiation papillopathy (373.45 Gy), radiation maculopathy (213.62 Gy), vitreous hemorrhage (1437.68 Gy), vascular occlusion (1080.93 Gy), nonproliferative retinopathy (1066.89 Gy), proliferative retinopathy (1590.71 Gy), exudative retinal detachment (1364.32 Gy), and rhegmatogenous retinal detachment (2265.54 Gy). Average biologically effective dose was higher in patients who developed radiation induced long term side effects than in the whole patient population except for radiation maculopathy. In spite of the small patient population and short-term follow-up, it is of interest to correlate the radiation induced effects and create a guideline for the improvement of the treatment of patients treated with LDR brachytherapy.

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## Introduction

The American Cancer Society's estimates for eye cancer in the United States for 2021 predict about 3320 cases of mostly melanoma of the orbit and potentially 400 deaths.<sup>1</sup> The current 5-year survival rate for people with eye cancer is 80%, and up to 85% if diagnosed at an early stage; however, the survival rates depend

on the size and location of the tumor, and the type of cancer. If the cancer metastasizes to other parts of the body, the 5-year survival rate dramatically drops to 15%.<sup>2</sup> Choroidal melanoma is the most common type of eye cancer. Ocular melanoma can metastasize via the blood, and 80 to 90% of the time, the liver is the affected organ. Among symptoms of eye cancer are light flashes, floating dark spots, watery eyes, blurry vision, loss of peripheral vision, eye displacement, and complete loss of the ability to see.

Ocular melanoma is a rare cancer believed to affect 5 to 6 people in a million. Therapeutic options include laser photocoagulation to heat and destroy cancer cells, chemotherapy injections, cryotherapy, radiation therapy, and enucleation. Radiation therapy

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